



The Shortage of Amoxicillin: An Escalating Public Health Crisis in Pediatrics Faced by Several Western Countries

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The antibiotic amoxicillin is usually recommended as a first-line treatment for many common infections affecting children, including otitis, pneumonia, sinusitis, and strep throat.¹ It is an effective, relatively inexpensive drug that is available in generic formulations.² For years, pediatricians have feared a major health crisis caused by the growing phenomenon of antibiotic resistance.³ Among the various causes, including unnecessary use of antibiotics in agriculture, poor sanitation practices, and poor infection control in hospitals and clinics, overuse and misuse of antibiotics in clinical practice have been recognized to be the main factors accelerating antibiotic resistance.⁴ However, antibiotic shortages have never been considered as a possible critical issue in public health and health care delivery. In particular, the unavailability of essential first-line antibiotics, such as amoxicillin, was considered a remote possibility.⁵ Contrary to these expectations, Authorities and Public Agencies in Europe and the US are now reporting shortages of amoxicillin,⁶⁻⁹ raising concerns that one of the most widely used antibiotics for the treatment of various pediatric conditions^{10,11} will be in short supply for the current winter season and for months to come.

This commentary, jointly authored by members of French, Israel, and Italian Societies of Pediatrics, in collaboration with the Confederation of European Pediatricians and the European Pediatric Association/Union of European National Pediatric Societies and Associations, briefly outlines the current public health crisis caused by the shortage of amoxicillin, which exposes children to life-threatening conditions.¹² The goal is to raise awareness among pediatricians, public health officials, and governments about the risks caused by prolonged drug shortages and particularly the unavailability of amoxicillin in Western countries and other areas of the world.¹³

Low Availability of Amoxicillin Causes Concern in Many Countries

More than 86% of European hospital pharmacists are experiencing difficulties in sourcing different types of medicines, and 66% report this as a daily or weekly problem. These shortages in Europe involve originator drugs in 51.8% of cases, and the percentage rises to approximately 60% if we include generic formulations. These alarming figures are presented in the 2022 report of the European Association of Hospital Pharmacists,¹³ which surveyed more than 600 hospital pharmacists from 36 different European countries.

The drugs most involved are cancer drugs (54.5%), emergency drugs (30.4%), cardiovascular drugs (30.4%), and anesthetics (26%).^{13,14} However, the greatest shortage is found among antimicrobial drugs, particularly antibiotics (56.7%). Among these, the shortage of amoxicillin raises particular alarm because of its importance in the treatment of bacterial infections across pediatrics. Amoxicillin accounts for approximately 70% of prescriptions for the management of acute otitis media.¹⁵

Some European countries in particular are suffering from the unavailability of amoxicillin. The situation is especially acute in France, where the Directorate General of the Ministry of Health has sounded the alarm about a severe shortage of amoxicillin, either alone or in combination with clavulanic acid, which are subject to severe supply shortages in this country.¹⁶ The unavailability of amoxicillin in its various formulations is considered particularly hazardous to children's health, with possible life-threatening consequences, by the French advisors to the Directorate, including the Pediatric Infectious Pathology Group of the French Society of Pediatrics, the French Association of Ambulatory Pediatrics, and the French Society of Infectious Pathology. They explain these shortages in amoxicillin by the very significant increase in antibiotic consumption coupled with difficulties on industrial production lines. During the coronavirus disease 2019 pandemic, the demand for amoxicillin had fallen sharply, leading to a reduction or even a stoppage of certain production lines, which have not recovered to their prepandemic production capacity. Moreover, if supply chain limitations are severe and/or prolonged, a domino effect of sorts is inevitable, with a shift in prescriptions to other oral compounds, such as cephalosporins, penicillin V, macrolides, adult forms of amoxicillin, and ceftriaxone. Stocks of these antibiotics are

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also not likely to be sufficient because they are much less prescribed, and they will be quickly depleted, perpetuating and exacerbating the shortage and increasing the risk of widespread antibiotic shortages. As noted in France, difficulties in the supply of pediatric forms of amoxicillin and amoxicillin clavulanate have the potential to create a major and unprecedented public health crisis with short- and long-term consequences. In Italy, the government drug agency has declared amoxicillin in its various formulations to be unavailable domestically, and the importation of equivalent drugs from abroad has been authorized to avoid a public health crisis.¹⁷ In Spain, the president of the local Council of Official Pharmacists Associations reported in November 2022 that pharmacies do not have enough stock to meet 50% of demand. He stressed that this is not due to an increase in consumption but rather from a lack of production because laboratories do not have the raw material needed for manufacturing.¹⁸ The Spanish Agency for Medicines and Health Products has reported in December 2022 shortages of 22 different amoxicillin drugs in tablets, powder, and hard capsules amid a wave of respiratory infections.¹⁹

In the US, the Food and Drug Administration, which lists drugs on its website when overall market demand is not met by product manufacturers, reported in November 2022 that amoxicillin oral powder for suspension used by children is in short supply.²⁰ Amoxicillin and clavulanate oral preparations in capsules and tablets also have limited availability in the US.²¹ Amoxicillin unavailability is reported in several other countries around the world, including Romania, Malaysia, Australia, and Canada.²²⁻²⁴

Main Causes of Shortages of Some Critical Drugs

An insufficient supply of drugs is considered one of the most significant problems in public health and medical care delivery.^{25,26} Two main factors can explain the drug shortages that periodically plague Europe: production shortages, which often recognize internal problems within pharmaceutical companies, and poor or temporary unavailability of distribution.²⁷ In Europe, non-European Union countries usually depend almost exclusively on the importation of drugs from larger producing countries, including Germany, Switzerland, and the US, or from non-European suppliers such as Brazil, India, and China. In the European Union, production-related shortages are mainly related to the importation of products between member states with different levels of product availability. Most of the production shortages are associated with various economic factors, including the cost of raw materials and their supply, low prices charged for final products, and the small size of the expected market for old drugs. Distribution unavailability, in contrast, is often caused by interstate market dynamics, which aim to control export (eg, restriction of products for export, or imported products released only after complex bureaucratic notification to the relevant authorities).²⁸ Sales between wholesalers operating in different markets with

different prices, the parallel trade, can often result in the unavailability of drugs essential to public health, as in the case of amoxicillin.

Conclusions

As has been noted in France, supply difficulties with pediatric forms of amoxicillin and amoxicillin clavulanate have the potential to create a serious and unprecedented public health crisis with short- and long-term consequences. Repeated lockdowns related to the coronavirus disease 2019 pandemic have contributed to supply difficulties for many drugs, including antibiotics. However, the risks associated with amoxicillin supply shortages appear not to have been sufficiently assessed, and the crisis we are facing today is serious and particularly dangerous for children's health. Further complicating the epidemiologic situation is the increasing reports of respiratory infections among children, caused by respiratory syncytial virus, which is quite common but can be dangerous for infants and children. The authors of this commentary, on behalf of their respective pediatric societies, wish to raise awareness among pediatricians and public health authorities nationally and internationally of the risks caused by prolonged unavailability of amoxicillin. Without rigorous measures to prevent shortages related to drug production and distribution, populations could face a postantibiotic era in which common infections and minor injuries can result in serious life-threatening situations.²⁹ ■

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